Review Article

# Polysaccharide-Derived Nanostructures: Engineering Strategies and Frontiers in Precision Nanomedicine

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## ABSTRACT

Nanotechnology has fundamentally revolutionized the biomedical landscape by providing high-precision solutions for diagnostics, therapeutics, and regenerative medicine. Among the myriads of available nanomaterials, carbohydrate-derived nanoparticles have emerged as premier candidates due to their superior biocompatibility, inherent biodegradability, low immunogenicity, and extensive structural versatility. This review provides a rigorous analysis of polysaccharide-based nanoparticles, delineating their primary synthetic methodologies, including ionic gelation, emulsification-solvent evaporation, and amphiphilic self-assembly, alongside critical physicochemical characterization techniques, such as dynamic light scattering and electron microscopy. A diverse library of polymers is examined, encompassing chitosan, hyaluronic acid (HA), alginate, and various microbial and marine-derived gums, each possessing unique functional moieties that facilitate specialized biological interactions. These nano constructs function as intelligent delivery vectors for pharmacological and genetic payloads, utilizing both passive accumulation via the enhanced permeation and retention (EPR) effect and active targeting through specific ligand-receptor affinities. Furthermore, their utility extends to advanced bioimaging as contrast agents, highly sensitive diagnostic biosensors, and bioactive scaffolds for tissue engineering and immunomodulation. Although challenges regarding industrial scalability and long-term toxicological profiles persist, the integration of stimuli-responsive mechanisms and theragnostic functionalities is projected to drive the next generation of personalized and efficacious nanomedicine.

## Introduction

Nanotechnology, defined as the precise manipulation of matter at the nanoscale (1-100 nm), has emerged as a transformative discipline with profound implications across the physical, chemical, and biological sciences. In the biomedical sector, the

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unique physicochemical properties of nanomaterials, such as their high surface-area-to-volume ratio, quantum effects, and tunable surface chemistry provide unprecedented opportunities for advanced diagnostics, targeted therapeutics, and regenerative medicine [1-3]. Among the vast array of nanoscale constructs currently under investigation, those derived from carbohydrates have garnered significant attention due to their inherent biocompatibility, biodegradability, low immunogenicity, and structural versatility [4].

Carbohydrates are ubiquitous in biological systems and fulfil a broad spectrum of physiological roles, including energy storage, structural support, and complex intercellular recognition events mediated by glycocalyx components on the cell surface [5]. Their polymeric nature, characterized by repeating monosaccharide units, facilitates the fabrication of diverse nano-architectures, including nanoparticles, nanofibers, and hydrogels, each offering distinct advantages for specific biomedical applications [6]. This review provides a comprehensive analysis of carbohydrate-derived nanoparticles, focusing on their synthetic methodologies, physicochemical characteristics, and multifaceted roles as carriers for drug and gene delivery, agents for bioimaging, platforms for diagnostics, and scaffolds for tissue engineering. By discussing both the synthesis of these materials and the mechanistic underpinnings of their biological interactions, this work aims to highlight their significant potential for clinical translation.

## *Polysaccharide-Based Nanomaterials: Historical Context and Contemporary Platforms*

The utilization of polysaccharide-based materials in medicine dates back to the use of alginates and chitosan for wound healing. However, their integration into nanomedicine has unlocked their full potential. Early research focused on natural polysaccharides due to their established safety profiles.

### *Chitosan*

Derived from the deacetylation of chitin, chitosan is a cationic polysaccharide widely studied for its mucoadhesive properties and ability to form nanoparticles through facile ionic gelation with polyanions such as tripolyphosphate (TPP) [7]. The positively charged amine groups of chitosan enable complexation with negatively charged therapeutic agents, including nucleic acids (siRNA, plasmid DNA), making it a leading non-viral vector for gene delivery [8].

### *Dextran*

A branched-chain polysaccharide, dextran, is known for its high-water solubility and low toxicity. It is commonly used as a surface coating for magnetic nanoparticles (*e.g.,* iron oxide nanoparticles) to improve their biocompatibility, reduce non-specific protein adsorption, and extend their circulation time for applications in magnetic resonance imaging (MRI) and magnetic hyperthermia [9].

### *HA*

As a major component of the extracellular matrix (ECM), HA is an anionic polysaccharide that is specifically recognized by CD44 receptors, which are often overexpressed on the surface of various cancer cells and inflamed tissues [10]. This specific ligand-receptor interaction provides a robust foundation for the development of actively targeted nanoparticles for precision medicine, particularly in oncology and inflammation-related diseases.

### *Cyclodextrins*

These cyclic oligosaccharides possess a unique toroidal structure with a hydrophobic inner cavity and a hydrophilic outer surface. This structural arrangement allows them to form host-guest inclusion complexes with poorly soluble hydrophobic drugs, thereby enhancing their solubility, stability, and bioavailability [11]. Cyclodextrin-based nanoparticles have been engineered to enable controlled and sustained drug release.

### *Alginate*

Alginate is a natural anionic polysaccharide derived primarily from brown seaweed, composed of *β-D*-mannuronic acid (M) and α-L-guluronic acid (G) residues linked by 1,4-glycosidic bonds. Its ability to form hydrogels and nanoparticles under mild conditions, typically in the presence of divalent cations like $Ca^{2+}$ (via an "egg-box" model of cross-linking), makes it highly attractive for sensitive biological cargo encapsulation [12,13]. Alginate nanoparticles are particularly useful for the controlled release of proteins, peptides, and cells, and have found applications in tissue engineering, drug delivery (especially for oral administration due to their stability in the gastrointestinal tract), and cell encapsulation [14]. Its biocompatibility and low toxicity contribute to its widespread adoption in various biomedical formulations.

### *Starch and Starch Derivatives*

Starch, a widely available and cost-effective polysaccharide, consists of two main components: amylose (a linear α-1,4-glucan) and amylopectin (a branched α-1,4/α-1,6-glucan). While native starch has limited solubility and mechanical strength, chemical modifications (*e.g.,* carboxymethylation, acetylation, and cross-linking) can yield derivatives with enhanced properties suitable for nanoparticle formation [15]. Starch-based nanoparticles can be fabricated through methods such as nanoprecipitation, emulsification, and enzymatic hydrolysis. These nanoparticles offer the potential for oral drug delivery, particularly for insoluble drugs, and as excipients in pharmaceutical formulations due to their biodegradability and non-toxicity [16]. Their versatility allows for tuneable degradation rates and drug release profiles depending on the degree and type of modification.

### *Heparin*

Heparin is a highly sulfated glycosaminoglycan (a linear polysaccharide) with a heterogeneous structure, primarily known for its anticoagulant properties [17]. Beyond anticoagulation, heparin exhibits diverse biological activities, including antithrombotic, anti-inflammatory, and anti-proliferative effects, as well as binding to a wide array of proteins (*e.g.,* growth factors, chemokines, and enzymes) [18]. This robust protein-binding capacity makes heparin an excellent candidate for the targeted delivery and controlled release of therapeutic proteins and growth factors. Heparin can be chemically modified (*e.g.,* thiolated heparin, low molecular weight heparin derivatives) and self-assembled or cross-linked to form nanoparticles. These nanoparticles have shown promise in wound healing, cardiovascular therapy, cancer treatment (by inhibiting angiogenesis and metastasis), and as platforms for the delivery of specific growth factors for tissue regeneration [19]. Its negative charge also allows for electrostatic interactions with cationic drugs or peptides.

### *Cellulose and Cellulose Nanocrystals (CNCs)*

Cellulose is the most abundant natural polymer on Earth, forming the primary structural component of plant cell walls. It is a linear polysaccharide composed of β-(1→4)-linked D-glucose units. While bulk cellulose is largely insoluble and rigid, advances in nanotechnology have enabled the isolation of CNCs and cellulose nanofibrils (CNFs), which are crystalline rod-like CNCs or highly entangled network (CNFs) nanostructures [19,20]. CNCs possess exceptionally high tensile strength, high surface area, excellent biocompatibility, and chemical modifiability due to the abundant hydroxyl groups. These properties make them highly attractive for biomedical applications, including drug delivery (*e.g.,* as drug carriers or stabilizers in emulsions), tissue engineering scaffolds, and diagnostic biosensors [21]. Their rod-like morphology can also facilitate anisotropic alignment, which is beneficial for certain tissue regeneration applications.

### *Pectin*

Pectin is a complex, branched polysaccharide primarily found in the cell walls of fruits and vegetables. Its main chain consists of linear α-(1→4)-linked D-galacturonic acid units, often interspersed with rhamnose residues and

substituted with neutral sugar side chains (*e.g.,* arabinan, galactan) [22]. Pectin's gelling properties, particularly in the presence of calcium ions or under acidic conditions, make it suitable for forming hydrogels and nanoparticles. It is highly biocompatible, biodegradable, and mucoadhesive, making pectin-based nanoparticles promising for oral drug delivery, especially for colon-targeted drug release due to its degradation by colonic microbiota [23]. Pectin nanoparticles have also been explored for vaccine delivery and as natural emulsifiers in various pharmaceutical formulations.

### *Pullulan*

Pullulan is a neutral, linear polysaccharide produced by the fungus Aureobasidium *pullulans*, composed of repeating maltotriose units linked by α-(1→6) glucosidic bonds [24]. It exhibits excellent water solubility, film-forming capabilities, and low viscosity, coupled with high biocompatibility and non-immunogenicity. These characteristics make pullulan an attractive polymer for nanoparticle formation, often through chemical modifications (*e.g.,* hydrophobic derivatization to enable self-assembly into micelles) or nanoprecipitation methods [25]. Pullulan-based nanoparticles have been extensively investigated for targeted drug delivery, particularly in cancer therapy, due to their ability to bind to asialoglycoprotein receptors (ASGPR) overexpressed on hepatocellular carcinoma cells. They are also explored as gene delivery vectors and for vaccine delivery due to their stability and ability to encapsulate various therapeutic agents [26].

### *Agar and Agarose*

Agar is a natural polysaccharide mixture extracted from red algae, primarily composed of agarose and agaropectin. Agarose is the main gelling component, a linear polymer of alternating D-galactose and 3,6-anhydro-L-galactose units linked by α-(1→3) and β-(1→4) glycosidic bonds [27]. Agarose's ability to form thermoreversible gels makes it highly valuable in biotechnology and biomedicine. Agarose nanoparticles can be prepared via emulsification, ionic gelation, or self-assembly methods, often involving chemical modification to introduce functional groups or improve solubility [28]. These nanoparticles offer excellent biocompatibility and low toxicity, making them suitable for encapsulating avarious therapeutic agents, including small molecules, proteins, and nucleic acids. They are also explored as scaffolds for cell encapsulation in tissue engineering, and as matrices for biosensors due to their robust mechanical properties and porous structure [29].

### *Guar Gum*

Guar gum is a non-ionic galactomannan polysaccharide derived from the seeds of the guar plant, *Cyamopsis tetragonoloba*. It consists of a linear chain of β-(1→4)-linked D-mannose units with α-(1→6)-linked D-galactose side units [30]. Guar gum is widely used in the food and pharmaceutical industries as a thickener, binder, and stabilizer due to its high viscosity and water solubility. Its chemical modification (*e.g.,* hydroxypropylation, carboxymethylation) allows for tunable physicochemical properties, enabling the formation of nanoparticles through various techniques such as solvent displacement, ionic gelation, or emulsion cross-linking [31]. Guar gum-based nanoparticles are particularly promising for oral drug delivery, especially for targeted delivery to the colon, as they are resistant to degradation in the upper gastrointestinal tract but are readily degraded by colonic microbiota. This makes them ideal for drugs treating inflammatory bowel diseases or for systemic absorption from the colon [32].

### *Chondroitin Sulfate (CS)*

CS is a linear anionic glycosaminoglycan, naturally occurring in the ECM of connective tissues, particularly cartilage. It is composed of repeating disaccharide units of D-glucuronic acid and N-acetylgalactosamine, with varying degrees and patterns of sulfation [33]. Similar to HA, CS can be specifically recognized by receptors (*e.g.,* CD44 and CD11b) overexpressed on various cell types, including cancer cells and immune cells, offering a pathway for active targeting [34]. CS can be used to form

nanoparticles either directly through self-assembly of hydrophobically modified CS, or by complexation with cationic polymers or drugs via electrostatic interactions. CS nanoparticles are being investigated for targeted drug delivery in cancer therapy, particularly for breast cancer and glioblastoma, as well as for applications in cartilage repair and regeneration due to their inherent biological role in these tissues [35]. Its anti-inflammatory and anti-angiogenic properties further enhance its therapeutic potential.

### *Gellan Gum*

Gellan gum is a water-soluble anionic polysaccharide produced by the bacterium *Sphingomonas elodea*. It is composed of repeating tetrasaccharide units of D-glucose, D-glucuronic acid, D-glucose, and L-rhamnose [36]. Gellan gum is widely recognized for its exceptional gelling properties, forming robust gels in the presence of mono- or divalent cations (*e.g.,* $Na^+$, $K^+$, $Ca^{2+}$, and $Mg^{2+}$) even at low concentrations [37]. This characteristic makes it highly valuable for the fabrication of nanoparticles via ionotropic gelation, emulsion cross-linking, or nanoprecipitation methods. Gellan gum nanoparticles are particularly suited for encapsulating sensitive biomolecules, such as proteins, growth factors, and drugs, providing sustained and controlled release. They have found promising applications in oral drug delivery, ocular drug delivery, and as scaffolds for tissue engineering, benefiting from their biocompatibility, biodegradability, and ability to form stable systems under physiological conditions [38].

### *Locust Bean Gum (LBG)*

LBG, also known as carob gum, is a neutral galactomannan polysaccharide extracted from the seeds of the carob tree (*Ceratonia siliqua*) [39]. Structurally, it consists of a linear backbone of β-(1→4)-linked D-mannopyranosyl units substituted with single α-(1→6)-linked D-galactopyranosyl residues. Compared to guar gum, LBG has a lower degree of galactose substitution, leading to less branching and allowing for stronger inter-chain associations, particularly in the presence of other hydrocolloids like xanthan gum, where it exhibits synergistic gelling effects [40]. LBG can be used to form nanoparticles through methods such as ionic gelation, nanoprecipitation, or emulsion-solvent evaporation, often after chemical modification to improve solubility or introduce specific functionalities. These nanoparticles are being explored for oral drug delivery, particularly for sustained release formulations, as well as for colon-specific drug targeting due to their resistance to enzymatic degradation in the upper gastrointestinal tract [41]. Its non-toxic and biocompatible nature makes it a suitable candidate for pharmaceutical and nutraceutical applications.

### *Xanthan Gum*

Xanthan gum is a high-molecular-weight anionic polysaccharide produced by the aerobic fermentation of the bacterium *Xanthomonas campestris*. Its primary structure consists of a cellulosic backbone (β-(1→4)-D-glucose units) with trisaccharide side chains (mannose-glucuronic acid-mannose) attached to alternate glucose residues [42]. These side chains are partially acetylated and contain pyruvate groups, contributing to their anionic character. Xanthan gum is well-known for its exceptional rheological properties, including high viscosity at low concentrations, shear-thinning behavior, and high stability across wide ranges of pH and temperature [43]. These properties make it an excellent material for formulating nanoparticles, often through complex coacervation with cationic polymers, emulsion techniques, or cross-linking. Xanthan gum nanoparticles are investigated for drug delivery systems, especially for sustained release formulations, mucoadhesive applications in ocular and oral drug delivery, and as effective stabilizers in pharmaceutical suspensions and emulsions [44]. Its stability in biological environments further enhances its utility.

### *Konjac glucomannan (KGM)*

KGM is a neutral, high-molecular-weight heteropolysaccharide obtained from the corm of the konjac plant (*Amorphophallus konjac*). It

primarily consists of β-(1→4)-linked D-glucose and D-mannose units in a ratio of approximately 1:1.6, with a small degree of acetylation [45]. KGM is renowned for its exceptional water-holding capacity, high viscosity, and thermo-reversible gelling properties, forming gels upon heating and cooling in alkaline conditions [46]. These properties make it an excellent material for fabricating nanoparticles through various methods, including ionic gelation, cross-linking, and self-assembly after chemical modification. KGM-based nanoparticles are highly biocompatible, biodegradable, and mucoadhesive, making them particularly suitable for oral drug delivery, sustained release formulations, and colon-targeted delivery. They are also investigated for tissue engineering applications as injectable hydrogels and for cell encapsulation [47].

### *Fucoidan*

Fucoidan is a complex sulfated polysaccharide found predominantly in the cell walls of various brown seaweeds and marine invertebrates. Its structure is highly variable depending on the source, but it primarily consists of a backbone of α-(1→3)-linked L-fucose residues, often with branching points at α-(1→4) linkages, and various degrees of sulfation [48]. Fucoidan exhibits a wide range of significant biological activities, including anticoagulant, anti-inflammatory, antiviral, antioxidant, and notably, anticancer properties [49]. The presence of sulfate groups contributes to its anionic nature, allowing for the formation of nanoparticles through electrostatic interactions with cationic polymers or drugs, or via self-assembly after hydrophobic modification. Fucoidan nanoparticles are being actively researched for targeted cancer therapy, particularly by leveraging their specific interactions with selectin receptors expressed on cancer cells and endothelial cells, as well as for their immunomodulatory effects and potential in drug delivery to treat various inflammatory conditions [50].

### *Gum Arabic (Acacia Gum)*

GA, also known as acacia gum, is a natural complex heteropolysaccharide obtained as an exudate from the *Acacia senegal* and *Acacia seyal* trees. It is a highly branched, anionic polysaccharide primarily composed of arabinose, galactose, rhamnose, and glucuronic acid, with a small protein component covalently attached to the carbohydrate backbone [51]. This unique arabinogalactan-protein structure gives GA excellent emulsifying, stabilizing, and film-forming properties. Gum Arabic nanoparticles can be formed through methods like nanoprecipitation, complex coacervation with proteins or cationic polymers, or direct emulsification. They are biocompatible, biodegradable, and have a long history of safe use in food and pharmaceuticals [52]. GA nanoparticles are explored as carriers for poorly soluble drugs, as stabilizers for nanoemulsions, and as potential delivery systems for vaccines and bioactive compounds, benefiting from their good encapsulation efficiency and stability in various media [53].

### *Inulin*

Inulin is a natural polysaccharide belonging to a group of fructans, primarily found in plants such as chicory, Jerusalem artichoke, and dahlia. It is a linear polymer predominantly composed of β-(2→1)-linked D-fructose units, often terminated by a glucose molecule at the reducing end [54]. Inulin is classified as a prebiotic fiber due to its resistance to digestion in the upper gastrointestinal tract and its fermentation by beneficial colonic bacteria. Its solubility and ability to form self-assembled nanoparticles (particularly after hydrophobic modification or under specific conditions) make it suitable for drug delivery applications [55]. Inulin-based nanoparticles are being explored for oral drug delivery, specifically for colon-targeted release to treat inflammatory bowel diseases or deliver vaccines. They are also investigated for their potential in regulating gut microbiota and enhancing the bioavailability of various compounds [56].

### *Chitin and Chitin Nanofibrils/Nanocrystals*

Chitin is the second most abundant natural polysaccharide after cellulose, serving as the main structural component of arthropod exoskeletons (*e.g.,* crustaceans and insects) and fungal cell walls [57]. It is a linear polymer of β-(1→4)-linked N-acetyl-D-glucosamine units. While chitin itself is highly insoluble and inert, it can be processed into nanoscale forms, specifically chitin nanofibrils (ChNFs) and chitin nanocrystals (ChNCs), which possess high mechanical strength, large surface area, and biocompatibility [58]. These nanoscale chitin derivatives can be used to form nanoparticles via methods such as electrospinning, nanoprecipitation, or top-down mechanical fibrillation, often incorporating other polymers or drugs. Chitin nanoparticles offer potential for drug delivery, wound healing, tissue engineering, and antimicrobial applications due to the inherent properties of chitin and its ability to interact with biological systems [59].

### *Dextrin and Maltodextrin*

Dextrin refers to a group of low-molecular-weight carbohydrates produced by the hydrolysis of starch (*e.g.,* maize, potato, and tapioca starch) through heat, acid, or enzymes. Maltodextrin is a specific type of dextrin with a dextrose equivalent (DE) value between 3 and 20. These polysaccharides are highly soluble in water, readily digestible, and generally recognized as safe (GRAS) [60]. Their branched structure and numerous hydroxyl groups provide sites for chemical modification, allowing for the formation of various nanoparticle constructs, often via methods such as nanoprecipitation, emulsion-crosslinking, or self-assembly of hydrophobically modified derivatives [61]. Dextrin/maltodextrin nanoparticles are versatile carriers for poorly soluble drugs, proteins, and nucleic acids. They are widely utilized in pharmaceutical formulations as excipients, as well as for controlled drug release, targeted delivery, and enhancing the stability and bioavailability of encapsulated compounds due to their non-toxic and biocompatible nature [62].

## Methodology: Synthetic Strategies and Characterization

The fabrication of carbohydrate nanoparticles requires robust and reproducible synthetic methodologies that preserve the structural integrity of both the polymer and the encapsulated cargo. The desired particle size, morphology, and application dictate the choice of method (Figure 1).

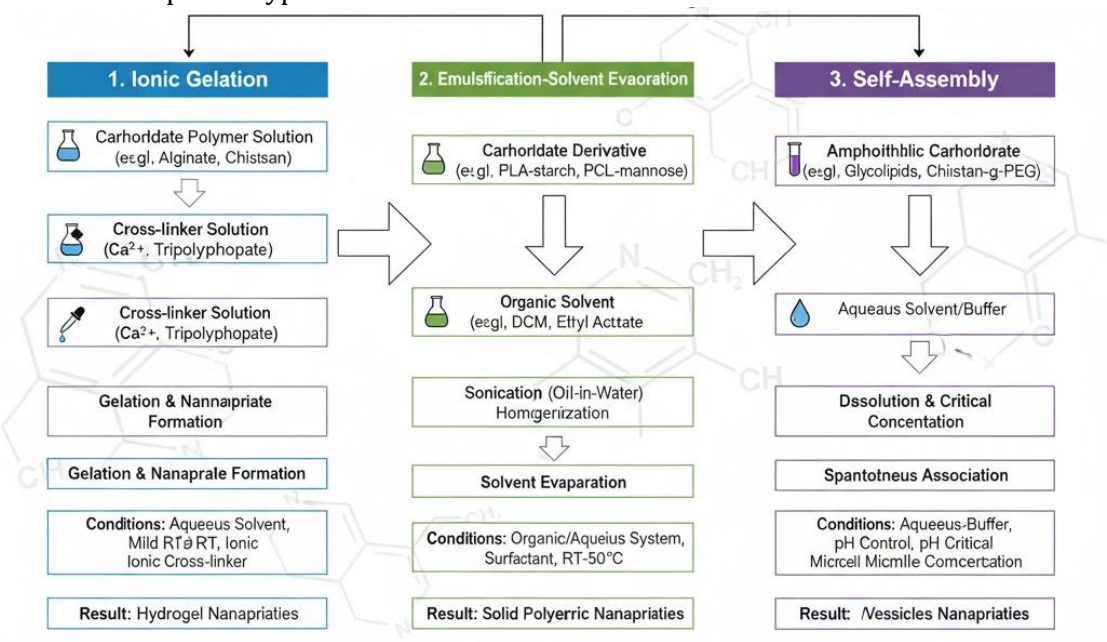



**Figure 1:** General synthetic pathways of carbohydrate-based nanoparticles

### *Ionic Gelation/Polyelectrolyte Complexation*

This method is predicated on the electrostatic interaction between oppositely charged polyelectrolytes. For instance, chitosan nanoparticles are synthesized by the controlled addition of a TPP solution to a chitosan solution, leading to the spontaneous formation of nanoparticles through polymer cross-linking [63]. This technique is favored for its simplicity, mild reaction conditions, and avoidance of harsh organic solvents.

### *Emulsification Solvent Evaporation/Diffusion*

This approach is typically used for hydrophobic polysaccharide [64] derivatives or for encapsulating hydrophobic drugs. The polymer and drug are dissolved in a volatile organic solvent (*e.g.,* dichloromethane and ethyl acetate) and emulsified in an aqueous phase containing a surfactant—subsequent evaporation of the organic solvent results in the precipitation of the polymer and the formation of nanoparticles [65]. The solvent diffusion method is a variant where the solvent is miscible with water.

### *Self-Assembly*

Amphiphilic polysaccharides, such as those modified with hydrophobic groups (*e.g.,* acyl chains), can spontaneously self-assemble into core-shell micellar structures in an aqueous environment above their critical micelle concentration (CMC) [66]. The hydrophobic core can effectively encapsulate therapeutic molecules, while the hydrophilic polysaccharide shell provides stability and biocompatibility.

### *Characterisation Techniques*

Accurate characterisation is fundamental to understanding the properties and behaviour of the nanoparticles (Figure 2).

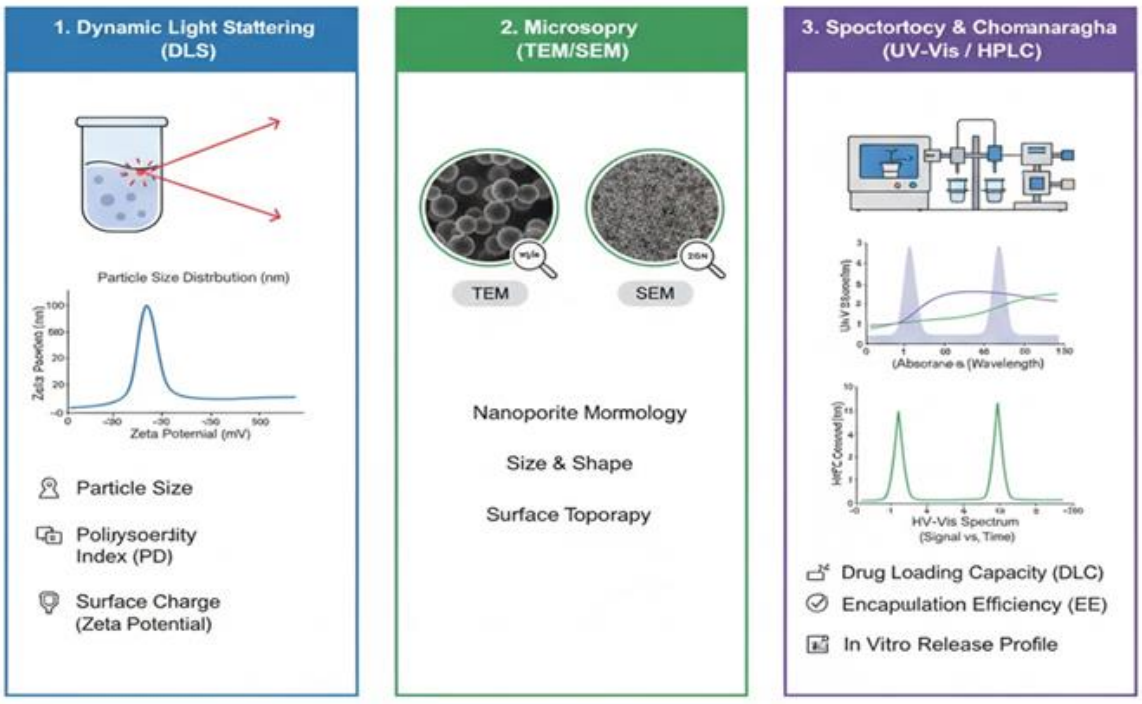



**Figure 2:** Characterisation Techniques

### *Particle Size and Zeta Potential*

Dynamic Light Scattering (DLS) is the standard technique to determine the hydrodynamic diameter and size distribution (polydispersity index) of the nanoparticles. Zeta potential measurement provides data on the surface charge, which is a key determinant of colloidal stability and cellular interaction [67].

### *Morphology*

The shape, size, and surface morphology are visualized using transmission electron microscopy (TEM) and scanning electron microscopy (SEM). TEM provides high-resolution images of the internal structure, while SEM reveals the surface topography.

### *Drug Encapsulation and Release*

The efficiency of drug encapsulation is quantified by separating the nanoparticles from the unencapsulated drug and measuring the drug concentration using techniques such as UV-Vis spectroscopy or high-performance liquid chromatography (HPLC). The *in vitro* release profile is determined by monitoring the drug concentration over time under simulated physiological conditions (*e.g.,* different pH values) [68].

## Results and Discussion

### *Current Advances in Carbohydrate Nanotechnology*

Biomedical Applications: A Mechanistic Analysis" refers to studying how a biological or biomedical system works at the level of cause-and-effect, rather than only observing outcomes. Such analysis helps explain how materials, drugs, signals, or models interact with cells, tissues, and disease processes, which is important for drug discovery, diagnostics, and personalized medicine. It connects mechanism with application: first understanding the biological process, then using that understanding to design better therapies, implants, sensors, or predictive models. This approach is valuable because it improves interpretability, supports safer design, and can reveal why a biomedical intervention works or fails.

All above mentioned compounds biomedical application are summarized in Table 1.

**Table 1**: Summary of polysaccharide-based nanomaterials and their biomedical functions

| | Polymer | Charge | Structural features | Primary synthesis method(s) | Key biomedical application(s) |
|---|---|---|---|---|---|
| 1 | Chitosan [7, 8] | Cationic | polysaccharide derived from the deacetylation of chitin, characterized by positively charged amine groups that enable mucoadhesion and electrostatic complexation | Ionic gelation | Gene delivery, mucoadhesive systems |
| 2 | Dextran [9] | Neutral | Branched, hydrophilic. polysaccharide utilized as a hydrophilic surface coating. | Surface coating, chemical modification | MRI contrast agents (SPIONs), protein adsorption reduction |
| 3 | Hyaluronic acid [10] | Anionic | glycosaminoglycan and major component of the ECM that specifically binds to CD44 receptors. | Self-assembly, chemical conjugation | Active cancer targeting (CD44 receptor), oncology |
| 4 | Cyclodextrins [11] | Neutral | Cyclic oligosaccharides possessing a unique toroidal (donut-shaped) structure with a hydrophobic inner cavity | Host-guest inclusion complexation | Solubility enhancement of hydrophobic drugs |

| | | | | | |
|---|---|---|---|---|---|
| | | | and a hydrophilic outer surface. | | |
| 5 | Alginate [12-14] | Anionic | polysaccharide composed of M and G residues linked by 1,4-glycosidic bonds, which forms hydrogels via the "egg-box" model | Ionic/Ionotropic gelation ($Ca^{2+}$) | Protein/peptide delivery, cell encapsulation |
| 6 | Starch [15,16] | Neutral | Consists of two main linear/branched glucans, glucose polymers: amylose (linear α-(1→4)-glucan) and amylopectin (branched α-(1→4)-/α-(1→6)-glucan). | Nanoprecipitation, emulsification | Oral drug delivery, pharmaceutical excipients |
| 7 | Heparin [17-19] | Anionic | Sulfated GAG. Cyclic oligosaccharides possessing a unique toroidal (donut-shaped) structure with a hydrophobic inner cavity and a hydrophilic outer surface, facilitating host-guest inclusion complexes. | Self-assembly, cross-linking | Growth factor delivery, cardiovascular therapy |
| 8 | Cellulose (CNCs) [19-21] | Neutral | linear polysaccharides composed of β-(1→4)-linked D-glucose units that form crystalline rod-like | Mechanical fibrillation, nanoprecipitation | Tissue engineering scaffolds, drug carriers |
| 9 | Pectin [22, 23] | Anionic | Branched acidic polysaccharide rich in galacturonic acid units; it is characterized by "smooth" homogalacturonan regions and "hairy" rhamnogalacturonan regions | Ionotropic gelation | Colon-targeted delivery, vaccine delivery |
| 10 | Pullulan [24-26] | Neutral | Polysaccharide. linear water-soluble glucan made of maltotriose units connected by α-(1→6) glycosidic bonds, which provides high flexibility and distinct oxygen-barrier properties | Nanoprecipitation, self-assembly | Liver targeting (ASGPR), cancer therapy |
| 11 | Agarose [27-29] | Neutral | linear polymer composed of alternating D-galactose | Emulsification, ionic gelation | Biosensors, tissue engineering scaffolds |

| | | | | | |
|---|---|---|---|---|---|
| | | | and 3,6-anhydro-L-galactopyranose units, known for forming thermally reversible double-helix structures that lead to rigid gels | | |
| 12 | Guar Gum [30-32] | Non-ionic | galactomannan consisting of a linear chain of β-(1→4)-linked D-mannose units with α-(1→6)-linked D-galactose side units. | Solvent displacement, ionic gelation | Colon-specific drug targeting |
| 13 | CS [33-35] | Anionic | glycosaminoglycan composed of repeating disaccharide units of D-glucuronic acid and N-acetylgalactosamine. | Complexation, self-assembly | Cartilage repair, active tumor targeting |
| 14 | Gellan Gum [36-38] | Anionic | polysaccharide composed of repeating tetrasaccharide units of D-glucose, D-glucuronic acid, D-glucose, and L-rhamnose. | Ionotropic gelation | Ocular drug delivery, growth factor encapsulation |
| 15 | Locust bean Gum [39-41] | Neutral | linear backbone of β-(1→4)-linked D-mannopyranosyl units substituted with single α-(1→6)-linked D-galactopyranosyl residues. | Nanoprecipitation, emulsification | Sustained release, colon-specific targeting |
| 16 | Xanthan Gum [42-44] | Anionic | high-molecular-weight polysaccharide with a cellulosic: backbone β-(1→4)-D-glucose units and trisaccharide side chains (mannose-glucuronic acid-mannose) | Complex coacervation, cross-linking | Mucoadhesive oral/ocular delivery, stabilizers |
| 17 | KGM [45-47] | Neutral | High-molecular-weight heteropolysaccharide, β-(1→4)-linked D-glucose and D-mannose | Ionic gelation, self-assembly | Injectable hydrogels, colon-targeted delivery |
| 18 | Fucoidan [48-50] | Anionic | Complex sulfated polysaccharide, α-(1→3)-linked L-fucose backbone | Electrostatic interaction, self-assembly | Targeted cancer therapy, immunomodulation |
| 19 | Gum Arabic [51-53] | Anionic | Highly branched arabinogalactan-protein structure | Complex coacervation, nanoprecipitation | Stabilizers for nano-emulsions, vaccine delivery |

| | | | | | |
|---|---|---|---|---|---|
| 20 | Inulin [54-56] | Neutral | Linear fructan, β-(2→1)-linked D-fructose units | Self-assembly, chemical modification | Prebiotic regulation, colon-targeted release |
| 21 | Chitin (ChNFs) [57-59] | Neutral | Linear polymer of β-(1→4)-linked N-acetyl-D-glucosamine | Electrospinning, nanoprecipitation | Antimicrobial applications, wound healing |
| 22 | Dextrin [60-62] | Neutral | Low-molecular-weight branched carbohydrate from starch hydrolysis | Emulsion-crosslinking, self-assembly | Controlled release, protein/nucleic acid carriers |

### *Targeted Drug and Gene Delivery*

Two distinct mechanistic pathways govern the localization of carbohydrate nanoparticles to specific pathological sites:

### *Passive Targeting*

This mechanism relies on the EPR effect. Nanoparticles preferentially accumulate in neoplastic tissues due to the presence of fenestrated, "leaky" tumor vasculature, and a lack of functional lymphatic drainage, which prevents efficient clearance [69].

### *Active Targeting*

Precision delivery is achieved by functionalizing the nanoparticle surface with specific carbohydrate ligands. These ligands facilitate high-affinity binding to receptors overexpressed on target cells. A hallmark example is the interaction between HA and CD44 receptors in oncological therapy (Figure 3) [69-70].

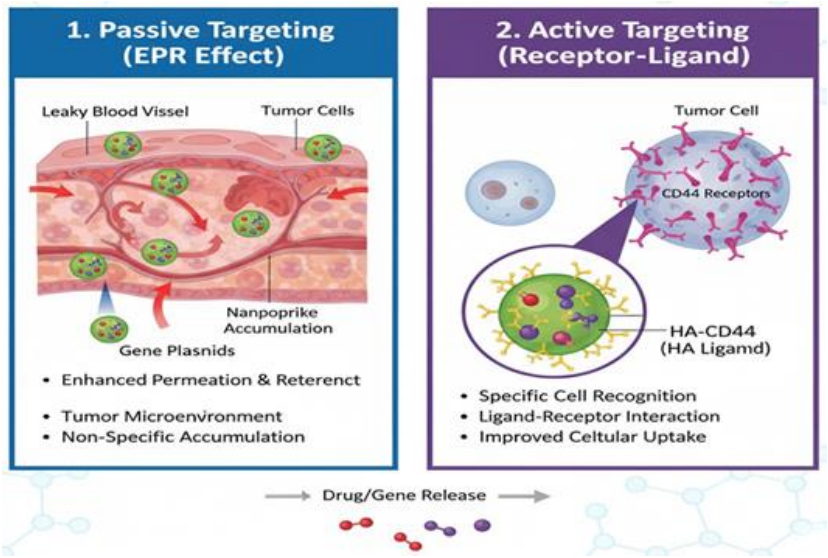



**Figure 3:** Mechanism of targeted drug and gene delivery

### *Case Study*

HA-conjugated nanoparticles encapsulating doxorubicin demonstrate superior tumor accumulation and enhanced therapeutic outcomes in breast cancer models compared to free drug administration [70].

In gene therapy, cationic polysaccharides such as chitosan are used to form stable electrostatic complexes with polyanionic nucleic acids (*e.g.,* siRNA or plasmid DNA). This complexation serves a dual purpose: protecting the genetic cargo from enzymatic degradation and facilitating transmembrane uptake.

### *Mechanism of Endosomal Escape*

A critical phase in gene delivery is the release of the cargo into the cytoplasm. Chitosan facilitates this via the proton sponge effect. The buffering capacity of chitosan leads to a massive influx of protons and water into the endosome, causing osmotic swelling and subsequent membrane rupture, thereby liberating the genetic material [71].

### *Bioimaging and Diagnostics*

Carbohydrate-derived nanoparticles serve as versatile, "intelligent" platforms in nanomedicine. Their primary utility stems from a combination of inherent biological safety, characterized by high biocompatibility and biodegradability, and the ability to minimize systemic toxicity while maximizing the therapeutic index (Figure 4).

### *MRI*

Dextran-coated superparamagnetic iron oxide nanoparticles (SPIONs) represent a clinically validated class of contrast agents. The dextran coating is crucial for maintaining colloidal stability, preventing opsonisation, and prolonging blood circulation time. This results in enhanced signal contrast in T2-weighted imaging, facilitating the high-resolution detection of tumors and inflammatory lesions [72].

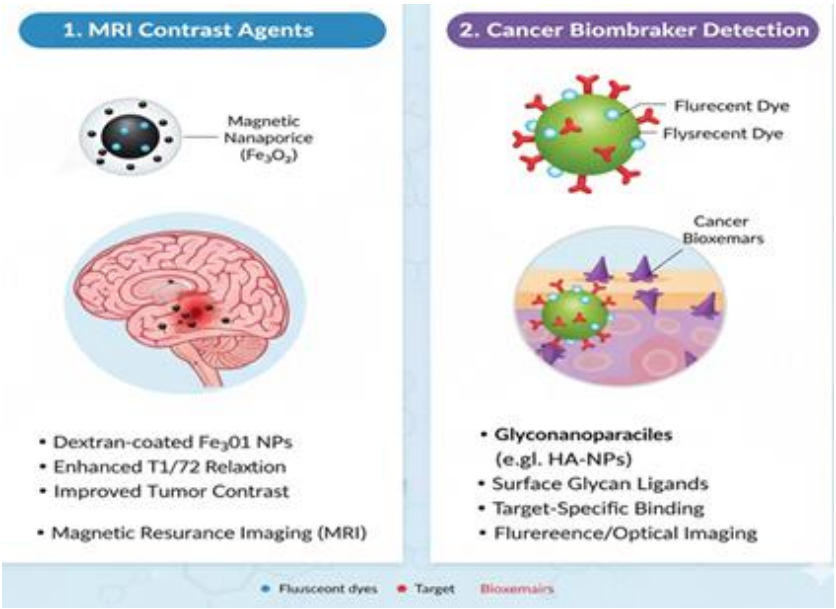



**Figure 4:** Role of polysaccharide nanoparticles in bioimaging and diagnostics

### *Fluorescence Imaging*

Polysaccharide platforms can be covalently conjugated to fluorophores or quantum dots. These systems allow for the real-time, longitudinal tracking of nanoparticle biodistribution, intracellular trafficking pathways, and drug release kinetics *in vivo* [73,74].

### *Molecular Diagnostics*

Glyconanoparticles are engineered to mimic the complex glycan topographies found on cell surfaces. These particles act as multivalent ligands, leveraging the cluster glycoside effect to dramatically increase binding avidity. This high-sensitivity binding is exploited for the detection of pathogens, toxins, and specific cancer biomarkers [74-76].

A significant advantage of using polysaccharide-based nanocarriers in bioimaging is their inherent biological safety, which helps to minimize the systemic toxicity typically

associated with raw contrast agents. However, it is important to clarify that while the carbohydrate carrier improves the safety profile by controlling biodistribution, the overall toxicity of the formulation remains dependent on the specific payload (such as heavy-metal-based contrast agents or fluorophores) and the stability of the entire nano construct *in vivo* [77].

### *Regenerative Medicine and Immunomodulation*

The structural similarity between natural polysaccharides and the ECM makes these materials ideal for regenerative applications [75].

### *Tissue Engineering Scaffolds*

Nanoparticles can be integrated within hydrogel matrices to function as localized drug depots [77]. These systems provide the controlled, sustained release of morphogens, such as bone morphogenetic proteins or cytokines to direct cellular proliferation and differentiation. For instance, alginate-based nanoparticles are used to deliver growth factors directly to wound sites to catalyze tissue repair [78] (Figure 5).

In the context of tissue engineering, polysaccharide nanoparticles are frequently integrated within macro-scale hydrogel systems. A clear distinction should be maintained between the two: while nanoparticles themselves are rarely used as the primary standalone scaffold, they serve a vital role as functional fillers or localized delivery depots embedded within a larger hydrogel matrix [78-80]. This hybrid approach enhances the mechanical integrity of the scaffold while providing a sustained release of growth factors or signaling molecules to the surrounding cellular environment [81].

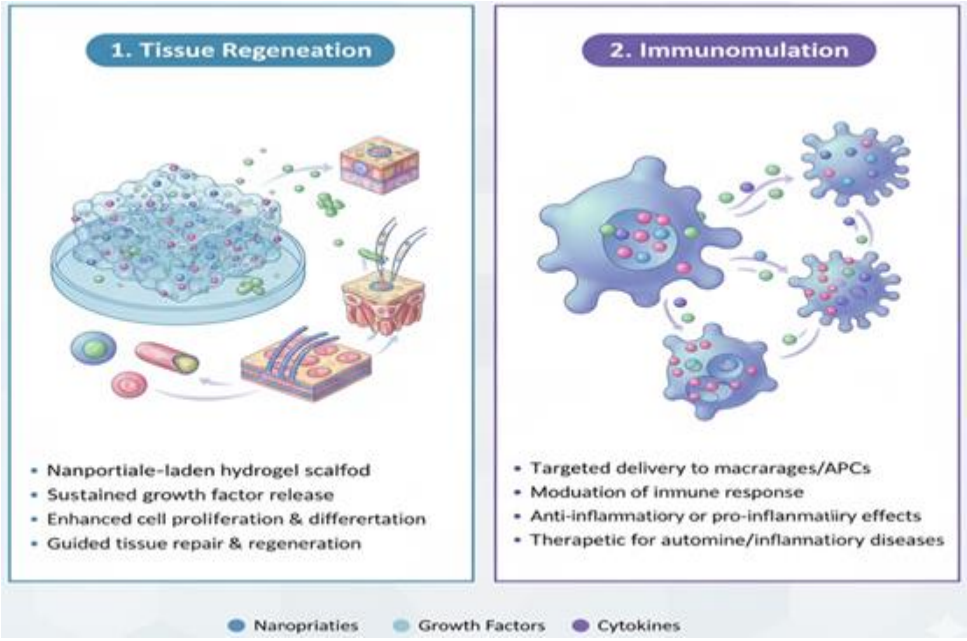



**Figure 5:** Regenerative medicine and immunomodulation applications

### *Immunomodulation and Vaccine Delivery*

Certain polysaccharides, particularly fungal-derived β-glucans, possess intrinsic immunomodulatory activity. These nanoparticles function as potent vaccine adjuvants by targeting antigen-presenting cells (APCs), such as dendritic cells and macrophages. By co-delivering antigens and stimulatory signals, they activate both innate and adaptive immune pathways, leading to robust and durable protective immunity [82,83].

### *Current Challenges: Scalability and Toxicity*

Despite the promising therapeutic potential of carbohydrate-derived nanoparticles, several critical hurdles must be cleared before widespread clinical translation can be realized [84]. A primary concern is industrial scalability; while laboratory-scale synthesis via methods such as ionic gelation or self-assembly is straightforward, maintaining batch-to-batch consistency and precise physicochemical control during large-scale manufacturing remains technically demanding. Furthermore, although these polysaccharides are inherently biocompatible, their long-term toxicological profiles require more rigorous investigation [85]. The metabolic fate of chemically modified derivatives and the potential for unintended immunogenic responses following chronic administration are not yet fully understood, necessitating extensive longitudinal safety studies to ensure patient safety. Addressing these gaps in manufacturing reproducibility and biological safety is essential for moving the next generation of polysaccharide nanomedicines from the benchtop to the clinic [86] (Table 2).

**Table 2:** Future trends in carbohydrate nanoparticles

| Innovation | Description | Biomedical potential |
|---|---|---|
| Stimuli-responsive systems | pH/redox/enzyme-triggered release | Controlled therapy |
| Theranostic nanoplatforms | Dual imaging + therapy | Personalized medicine |
| 3D nanoscaffolds | ECM-mimetic materials | Tissue regeneration |
| Hybrid nanocomposites | Carbohydrate + Metal/organic frameworks | Enhanced targeting and stability |

## Conclusion and Future Perspectives

The application of carbohydrate-derived nanoparticles in biomedicine is a testament to the power of integrating natural biomaterials with cutting-edge nanotechnological principles. Their inherent biocompatibility, biodegradability, and versatile structural chemistry make them an ideal platform for designing sophisticated and safe nanomedicines. While significant progress has been made, several challenges remain. The clinical translation of these nanomedicines requires addressing the scalability of synthesis, ensuring batch-to-batch consistency, and conducting rigorous long-term toxicity and immunogenicity studies.

Future research sholud explore several exciting frontiers in the development of carbohydrate-based nanoparticles. One key area is stimuli-responsive systems, which involve designing "smart" nanoparticles that can react to specific cues in their microenvironment, such as changes in pH, redox potential, or enzyme concentration. This responsiveness enables on-demand drug release, significantly enhancing therapeutic efficacy and minimizing off-target effects. Another promising avenue is the creation of theranostic platforms. These multifunctional nanoparticles would combine diagnostic imaging capabilities with therapeutic drug delivery, allowing for real-time monitoring of treatment responses and paving the way for truly personalized medicine. Researchers are also focusing on combination therapies, using carbohydrate nanoparticles to co-deliver multiple therapeutic agents, such as chemotherapy and gene therapy, to achieve synergistic effects and combat challenges such as drug resistance in complex diseases like cancer. Lastly, immunotherapy is a rapidly growing field where the immunomodulatory properties of carbohydrate nanoparticles are being harnessed to develop novel cancer vaccines and other immunotherapies. As the understanding of nanoscale bio-interactions continues to deepen, carbohydrate-based nanoparticles are set to play a pivotal role in shaping the future of healthcare, offering solutions that are more effective, safer, and highly personalized for individual patients.

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